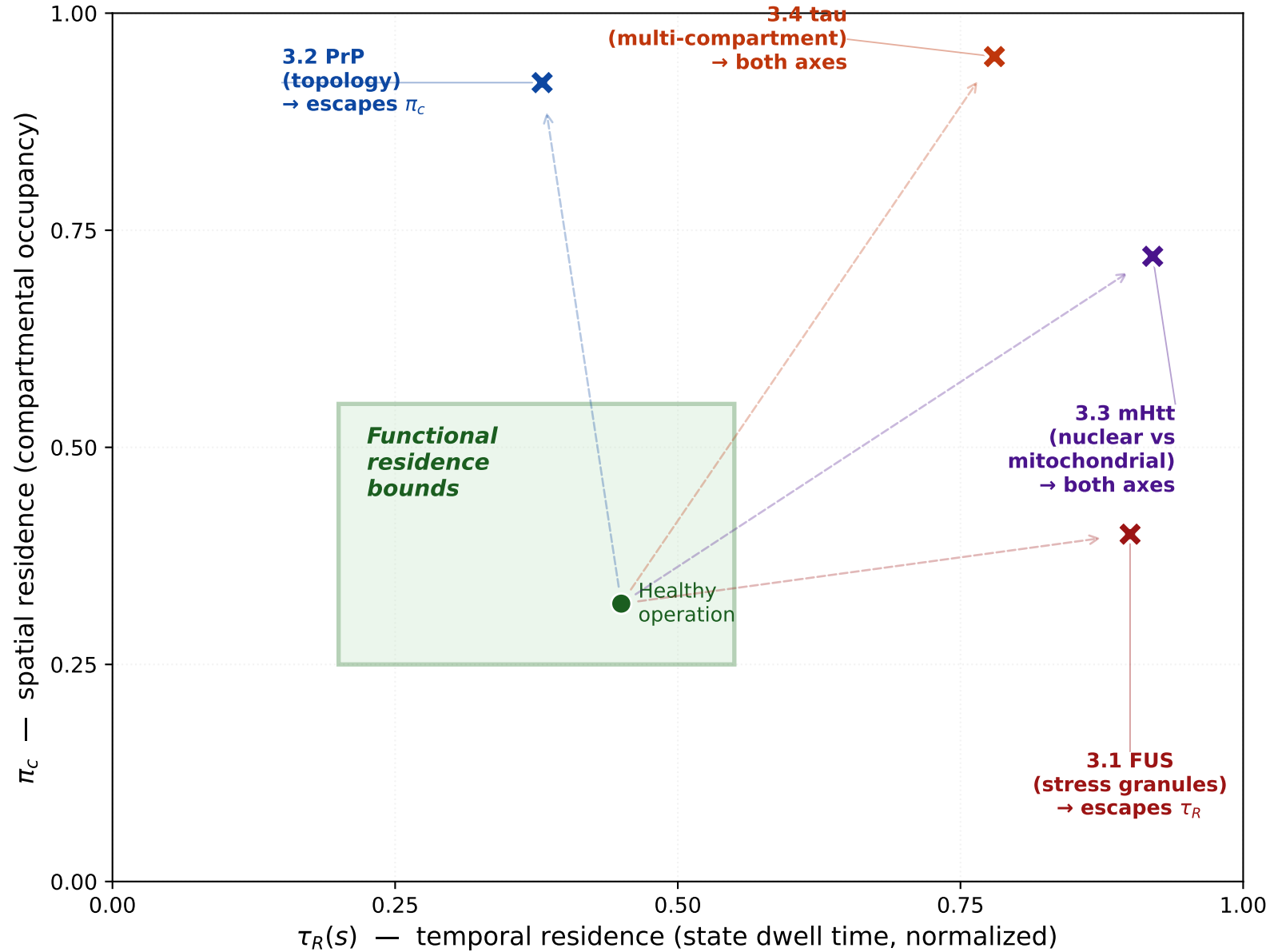


Figure 1. The two-axis residence observable $R(s, c) = \tau_R(s) \times \pi_c$ with four canonical disease cases (§3) demonstrating two-axis necessity



Pathology corresponds to $R(s, c)$ escaping the functional residence bounds on τ_R , π_c , or both. Pure-axis cases (3.1, 3.2) establish independent empirical content of each axis; both-axes cases (3.3, 3.4) establish that one-axis observables cannot resolve clinical heterogeneity within a single protein.